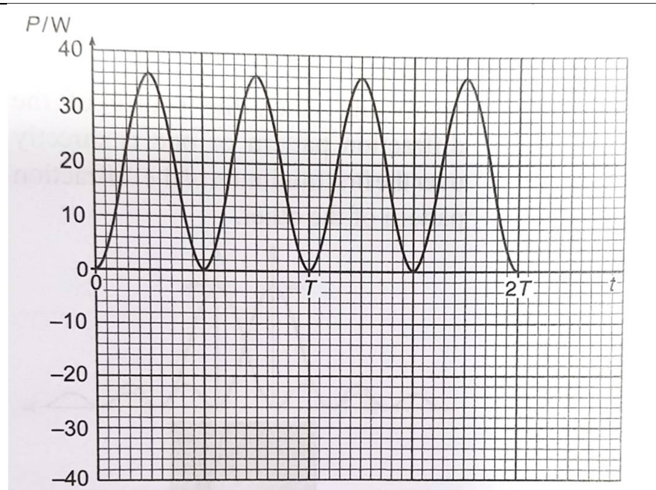
	$= 0.150\ \text{V}$
7(a)	The root-mean-square value of an alternating current is the equivalent value of a steady direct current that will dissipate heat at the same average rate as the alternating current in a given resistor.
7(b)(i)	r.m.s. potential difference $V_{\text{r.m.s.}} = \dfrac{V_{\text{peak}}}{\sqrt{2}}$ $= \dfrac{24}{\sqrt{2}}$ $= 16.97$ $\approx 17\ \text{V}$
7(b)(ii)	frequency

No.	Solution
	$f=\dfrac{\omega}{2\pi}$ $=\dfrac{440}{2\pi}$ $=70.03$ $\approx 70\text{ Hz}$
7(c)(i)	Mean power dissipated in heater $\langle P\rangle=\dfrac{V_{\text{r.m.s.}}^2}{R}$ $=\dfrac{17^2}{16}$ $=18.06$ $\approx 18\text{ W}$
7(c)(ii)	
8(a)(i)	A photon is a quantum of electromagnetic radiation with a discrete amount of